

Arun Kumar Thiyagarajan

Aspiring AI & Machine Learning Engineer | Python | Machine Learning | Data Analysis

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PROFESSIONAL SUMMARY

Recent B.Tech graduate in Artificial Intelligence and Data Science with strong foundations in Python, Machine Learning, and Data Analysis. Built projects including Customer Churn Prediction and AI PDF Question Answering System using Scikit-Learn, Flask, and Streamlit. Seeking an entry-level opportunity to apply and enhance my technical skills.

EDUCATION

United Institute Of Technology 11/2022 – 05/2026
Bachelor of Technology (B.Tech) in Artificial Intelligence and Data Science

IIT(Patna) 02/2026 – 06/2026
Certification program in Artificial Intelligence and Machine Learning Essentials

SKILLS

Programming Languages: Python, SQL

Data Preparation & Analysis: Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Data Visualization

ML Modeling & Evaluation: Regression, Classification, Model Training, Model Evaluation (Accuracy, Precision, Recall, F1-Score) |

Frameworks & Libraries: Scikit-learn, Pandas, NumPy, Matplotlib |

Tools & Workflow: Git, GitHub, Jupyter Notebook, Google Colab, Visual Studio Code | **Soft Skills:** Stakeholder communication, Team collaboration, Analytical thinking.

PROJECTS

AI PDF QUESTION ANSWERING SYSTEM | Python | Streamlit | LangChain | FAISS | Google Gemini

- Developed an AI-powered application that answers user questions based on uploaded PDF documents using a Retrieval-Augmented Generation (RAG) pipeline.
- Extracted PDF content using PyPDF, split documents into smaller chunks, and generated vector embeddings for efficient document retrieval.
- Executed FAISS vector search with LangChain to retrieve the most relevant document sections before generating responses.
- Merged Google Gemini to generate context-aware answers based only on the retrieved PDF content.
- Built an interactive Streamlit interface that enables users to upload PDF files and ask questions in real time.

AI RESUME PARSER | Python | Streamlit | Google Gemini | PyPDF

- Expanded an AI-powered Resume Parser that extracts structured information from PDF resumes using Large Language Models.
- Built an interactive Streamlit interface to upload resumes and automatically process PDF content using PyPDF.
- Blended Google Gemini API to identify and extract candidate details such as personal information, education, skills, experience, and projects into structured JSON format.
- Enforced environment variable management using dotenv for secure API key handling and improved application security.
- Designed a clean and user-friendly workflow for automated resume screening, reducing manual data extraction effort.

Bank Customer Churn Prediction | Machine Learning

- Improved a machine learning-based web application to predict whether a bank customer is likely to churn based on customer demographics and account details.
- Performed data preprocessing, feature encoding, and exploratory data analysis (EDA) on customer data.
- Trained and evaluated multiple machine learning models to classify customer churn and selected the best-performing model.
- Built an interactive web application using Flask to collect customer information and display real-time churn predictions.
- Executed categorical feature encoding and integrated the trained model using Joblib for deployment.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Flask, HTML, CSS, Joblib

CERTIFICATES

Artificial Intelligence & Data Scientist — Tamil Nadu Academy of Construction (2026)

Applied Artificial Intelligence and Machine Learning Essentials — IIT Patna & Masai School (2026)

Introduction to Generative AI — Google Cloud (2026)

IBM Virtual Internship — Web, Mobile Development and Marketing (2026)